Sahaj Bhadja

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Summary

Full-stack Software Engineer pursuing an MSCS at USC with 2+ years building backend data pipelines, ETL workflows, and monitoring dashboards in Python, SQL, and React across distributed systems and cloud platforms, with strong focus on data integrity, performance, and production reliability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Designed and maintained **backend data pipelines and ETL workflows** across FastAPI, PostgreSQL, SQL Server, and cloud storage, processing 5K+ monthly jobs with incremental loads and 99%+ production reliability.
- Resolved **data integrity issues** across schemas and timestamps with validation and reconciliation checks, optimizing SQL queries and indexing to cut dashboard latency from 1.2s to 350ms.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Built **Python and Flask backend services** with REST APIs and a React dashboard for 150+ users, handling 10K+ daily requests with CI/CD-driven releases and stable performance under load.
- Engineered a **concurrent batch-processing pipeline** with secure ingestion and AWS RDS, validating throughput against production-scale datasets and reducing turnaround time **60%**.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Led **ETL pipeline redesign** using AWS Glue and S3, boosting pipeline efficiency **68%**, and optimized large-scale API integrations with OOP (Java), improving response times **30%**.
- Built automated **load and integration testing** across Windows, Linux, and macOS with Jenkins, PyTest, and Mockito, maintaining 20+ test suites and validating performance at scale.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **full-stack MERN dashboards and REST APIs** across eight microservices syncing 2M+ records, with real-time WebSocket updates and role-based access control.
- Built **real-time monitoring** with Sentry, Azure Monitor, and Kibana, automating Docker/Jenkins CI/CD and cutting deployment time **50%** with DevSecOps checks (Snyk, OWASP).

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Built a **distributed data-pipeline framework** using Databricks, Spark, and Flask APIs, enabling real-time analytics on 50K+ records with sub-200ms responses.
- Integrated and tested **payment and webhook APIs** with token-based auth, reducing manual handling **70%** and documenting integration architecture for cross-team alignment.

Skills

Languages: Python, SQL, Java, JavaScript/TypeScript, C++, Bash
Data & ETL: ETL Pipelines, Data Integration, Incremental Loads, Data Validation, Reconciliation, Data Warehousing, Distributed Systems
Databases: PostgreSQL, SQL Server, MySQL, Azure SQL, Databricks, MongoDB
Backend & APIs: FastAPI, Flask, Node.js, REST APIs, Microservices, API Testing (Postman)
Frontend: React.js, Next.js, TailwindCSS, Data-Driven Dashboards, HTML/CSS
Cloud & DevOps: AWS (Glue, S3, RDS), Azure, Docker, Kubernetes, CI/CD, Git, Monitoring (Sentry, Kibana)

Projects

Big-Brain Live | *GCP, PostgreSQL, REST APIs*

- Engineered a **cloud data ingestion and indexing pipeline** on GCP processing 50K+ records, implementing incremental updates, metadata persistence, and tuning to reduce retrieval latency **40%**.

Linguistix | *Python, AWS (ECS/Fargate), Docker, CI/CD*

- Developed a **containerized RESTful data-processing API** on AWS (CodePipeline, ECS/Fargate) with automated CI/CD, validation checks, and 87% accuracy at production-grade reliability.

FinNexus Live | *Next.js, Kubernetes, WebSockets, MongoDB*

- Built a **real-time analytics platform** on Kubernetes for 50K+ users and 1M+ transactions with WebSocket updates, token-based access, and CI/CD-backed deployment.